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Question #01

every group of a's must come in multiple of 3

Regular Expression : $(b^*a^3a^*)^*b^*$ → $aaa, ba^3a, aaab, \dots$

Question #02

General Pattern:

$(a+b)^*S_1(a+b)^* + (a+b)^*S_2(a+b)^* + (a+b)^*S_3(a+b)^* + (a+b)^*S_4(a+b)^*$

Question #03

(a) Strings with exactly 3 b's

Regular Expression : $a^*ba^*ba^*ba^*$ → $bbb, abbb, bbba, a^2bba, \dots$

(b) Strings with exactly 2 or 3 b's

Regular Expression : $a^*ba^*ba^* + a^*ba^*ba^*ba^*$

2 b's → bb, abb, bab, a^2bb

3 b's → $bbb, abbb, bbba$

Question #04

(a) strings ending with double letter

Regular Expression : $(a+b)^*(aa+bb)$ → $aa, bb, baa, abb, babb, \dots$

(b) strings containing exactly one double letter

Regular Expression : $(a+b)^*(aa+bb)(a+b)^*$ → $aa, baa, abb, abba, babb, \dots$

Question #05

String where b never tripples

Regular Expression : $(a+ba+bba)^*$ → $a, ab, bb, bab, abb, \dots$

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Question #06

Contains aaa or bbb

Regular Expression: $(a+b)^* aaa (a+b)^* + (a+b)^* bbb (a+b)^*$

$a's \xrightarrow{aaa} aaa, baaa, aaab$

$b's \xrightarrow{bbb} bbb, abbb, bbba$

Question #07

(a) strings with no substring "ab"

Regular Expression: $b^* a^*$ $\rightarrow \epsilon, b, bb, ba, bba, bbaa, \dots$

(b) All strings that do not have both substrings bba and abb .

Concept:

L_1 : strings without bba $(a+b)^* - (a+b)^* bba (a+b)^*$

L_2 : strings without abb $(a+b)^* - (a+b)^* abb (a+b)^*$

$$L = L_1 \cup L_2$$

Regular Expression: $((a+b)^* - (a+b)^* bba (a+b)^*) + ((a+b)^* - (a+b)^* abb (a+b)^*)$

$\rightarrow abb, bba, abab, bbab, abba, babb, \dots$

Question #08

Number of a 's divisible by 3

$(b^* aaa)^* b^*$ $\rightarrow aaa, aabaaab, baaa,$

$baaaabaaa, \dots$

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Question # 09

(a) b's appear in odd groups

$b(bb)^*$
1, 3, 5, ...

~~$(a^* b (a^* bb)^* a^*)^*$~~

Regular Expression: $(a^* b (bb)^* a^*)^*$ → b, ab, ba, bbb, abbb, bbba, ...

(b) Even number of a's AND odd number of b's

even a's = $(aa)^*$

odd b's = $b(bb)^*$

Regular Expression: $(aa)^* b (bb)^* (aa)^*$ → b, bbba, aab, aabbb, ...

(c) Even number of a's and even number of b's

even a's = $(aa)^*$

even b's = $(bb)^*$

Regular Expression:

$(aa + bb + abab + baba)^*$ → 1, aa, bb, abab, baba, aabb, ...

(a) Question # 10

$RE_1 \longrightarrow (a+b)^* a (a+b)^* b (a+b)^*$

Language of RE_1 ab, aab, abb, abab, baab, aaabb, aba

$RE_2 \longrightarrow (a+b)^* ab (a+b)^*$

Language of RE_2 ab, aab, abb, abab, baab, aaabb

Both languages have strings containing substring "ab"

$(a+b)^* a (a+b)^* b (a+b)^* = (a+b)^* ab (a+b)^*$

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(b)

$$RE_1 \longrightarrow (a+b)^* ab (a+b)^* + b^* a^*$$

- part 1: $(a+b)^* ab (a+b)^*$ contain **ab**

Language: $ab, aab, abb, abab, baab, \dots$

- part 2: $b^* a^*$ **donot contain ab**

Language: $\Lambda, b, bb, a, aa, ba, bba, bbaa, \dots$

Language of RE_1 every possible string over a, b .

$$L_1 = (a+b)^*$$

$$RE_2 \longrightarrow (a+b)^*$$

Language of RE_2 every possible strings over a, b

$$L = (a+b)^*$$

~~$L_1 = L_2$~~

$$L_1 = L_2 = (a+b)^*$$

(c)

$$\text{Expression: } (a+b)^* ab [(a+b)^* ab (a+b)^* + b^* a^*] + b^* a^*$$

Already proved in part b

$$\text{Final Expression: } (a+b)^*$$

all possible strings over a, b

$\rightarrow \Lambda, a, b, ab, ba, aa, bb, aba, bab, aabb, \dots$

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Question #11 (Check RE Equivalence)

(a) $(ab)^*a$ and $a(ba)^*$

- strings of first : $a, aba, ababa, abababa, \dots$
- strings of second : $a, aba, ababa, abababa, \dots$

Equivalent

(b) $(a^*b)^*$ and $(a+b)^*$

- $a^* \rightarrow$ any number of a 's, and with $a \rightarrow$ repeat all strings over a, b
- $(a+b)^* \rightarrow$ all strings over a, b

Equivalent

(c) $(a^* + b^*)^*$ and $(a+b)^*$

- $a^* \rightarrow$ all a 's, $b^* \rightarrow$ all b 's $\rightarrow$ repeat all possible strings over a, b
- $(a+b)^* \rightarrow$ all string over a, b

Equivalent

(d) ϵ^* and ϵ

- As $\epsilon^* = \epsilon$

Equivalent

(e) $(a^*b)^*a$ and $a^*(ba^*)^*$

- Both $\rightarrow$ all strings over a, b with any number of a 's
- Language $\rightarrow \epsilon, a, aa, b, ab, aab, \dots$

Equivalent

(f) $(a^*bbb)^*a^*$ and $a^*(bbba^*)^*$

- Both $\rightarrow b$'s appear only in group of 3, any number of a 's
- Language $\rightarrow \epsilon, a, aaa, bbb, aaabbb, bbbbaaa, \dots$

Equivalent

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(g) $((a+bb)^*aa)^*$ and $\epsilon + (a+bb)^*aa$

First RE $\rightarrow \Lambda, aa, aaaa, bbbaa, aabaa, bbaaaa, \dots$

whole can repeat

Second RE $\rightarrow \epsilon$ or one string ends in aa

$\Lambda, aa, bbaa, aabaa, abbbaa, \dots$

only one block, not repeated

Not Equivalent

(h) $(aa)^*(\epsilon + a)$ and a^*

• First: even number of a's and option one extra a

• Second: $a, aa, aaa, aaaa, \dots$

Equivalent

(i) $a(ab)^*(\epsilon + a)b + b$ and a^*b

• First: $b, ab, aab, abab, abaab, \dots$

• Second: $b, ab, aab, aaab, aqaab, \dots$

Not Equivalent

(j) $a(ba + a^*ba)b$ and $aa^*b(aa^*b)^*$

• First: $ab, aab, aabab, aqaab, abab, aabab, \dots$

• Second: $ab, aab, aqaab, aqaab, abab, aabab, \dots$

Equivalent

(k) $\Lambda + a(a+b)^* + (a+b)^*aa(a+b)^*$ and $(b^*)a^*ab^*)^*$

• First: $\Lambda, a, ab, aba, abb, aa, baa, bbaa, babaa, \dots$

• Second: $\Lambda, a, ab, ba, aba, bab, aa, baa, baba, \dots$

Not Equivalent

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Question #12

(a) $(a+b)^* acbbb$

English: All strings over alphabet that end with acbbb

Language: acbbb, aacbbb, bacbbb, ...

(b) $(\overset{a}{a} + bbb)^*$

English: All string made by repeating a or bbb any number of times

Language: Λ , a, bbb, aa, abbb, bbbba, ...

(c) $(aaa)^* (bbb)^*$

English: All strings having any number of aaa blocks followed by any number of bbb.

Language: Λ , aaa, bbb, aaabbb, aaaaaabbbbbbb, ...

(d) $b(bbb)^* a(aa)^*$

English: start with one b, have any number of bbb, one a, any number of aa

Language: ba, baa, baaa, bbbba, ...

(e) $b(bbb)^* a(aa)^* b(bb)^* (aaa)^*$

English: start with b, have groups of bbb, one a, groups of aa, one b, groups of bb, groups of aaa

Language: bab, babaa, bbbhabb, ...

(f) $((a+b)a)^*$

English: All strings made by repeating 2-letter block whose 2nd letter is always a.

Language: Λ , aa, ba, aaba, baaa, baba, ...